Fusing LLM Semantic Repair with Genetic Programming for Robust Alpha Discovery

G.S.K. Raghav Anand ▪ NIT India ▪ IEEE Access manuscript

*Vanilla quant models break out-of-sample. A regime-aware GP + local-LLM repair loop fixes that — S&P 500 Sharpe decay cut from −35% **to** −7%.*

10.6% net return (peak) | **0.65** net Sharpe | **−7%** S&P decay (was −35%) | **+18%** over quant ceiling

NSE Nifty 50 2002–2025 (23 yrs) ▪ validated on S&P 500 + SSE ▪ costs 20+10 bps, cap INR 50 Cr ▪ local Llama-3.2-3B + Gemma-2-2B, $0 inference

SYSTEM ARCHITECTURE — read left to right (the core of this poster)

0 DATA — NSE 50
2002–2025 daily OHLCV + PE / ROE / EPS / BV
S&P 500 + SSE for test
What goes in

1 AFM (slow) — Macro anchor
FinBERT sentiment + fundamentals
Bull / bear prior
stops GP chasing noise

2 HMM (fast) — 2 volatility states
calm vs turbulent
Baum-Welch + Viterbi
Which regime now?
routes to specialist

3 GP SEARCH — Evolves formulas
pop 500, depth max 6
Sharpe minus bloat fitness
Invents alphas

4 LLM REPAIR — Judge, then fix
Llama scores 1–5
Gemma rewrites S below 3
Repair, not reject

~~repaired tree recompiled, back~~ into evolution (zero attrition)

OUTPUT: regime alphas x (AFM prior + HMM state) = dollar-neutral top/bottom-15% portfolio, monthly rebalance

Follow the five boxes once and you understand the paper. Green loop is the novelty: old hybrids deleted bad trees; this repairs them.

1 Problem in 30 seconds

- Quant factors **decay**: top-1000 anomalies lose about 32% after publication; GP memorises noise.
- Old LLM+GP hybrids **delete** bad formulas and lose diversity.
- Fix: **repair, don't reject** — rewrite only the truly nonsensical (score 2 or less).

2 What each layer does

- **AFM (slow):** news mood + PE/ROE/EPS/BV gives a bull/bear prior, an economic compass.
- **HMM (fast):** 2-state volatility filter. Calm goes to trend formulas; turbulence to reversals + quality.
- **GP (inventor):** 500 trees, depth max 6, Sharpe minus complexity fitness.
- **LLM (editor):** Llama rates economics 1–5; Gemma rewrites failures from a closed vocabulary.
- **Settings:** AFM FinBERT threshold $\tau = 0.5$; HMM 2-state Baum–Welch + Viterbi; GP pop 500, depth max 6; edit list $L \leq 5$; dollar-neutral top/bottom-15%, monthly rebalance.

2b Loop + stability (how it runs, why it holds)

GP loop: AFM prior + HMM state → fitness (Sharpe − complexity) → Llama judge (1–5) → Gemma repair ($S \leq 2$, closed vocab, $L \leq 5$) → recompile, back into evolution. Zero attrition: every repair compiles.



Rolling-window Sharpe stays positive (0.855) in turbulent 2017–20 where momentum + trend turn negative. Regime specialists, not one average model.

3 Does it work? Each layer adds return (NSE)



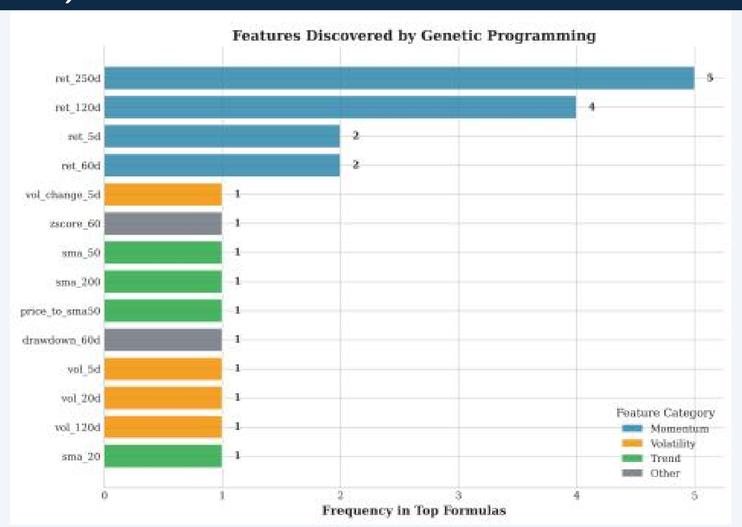
Ablation, NSE net: Vanilla 1.62% → +AFM 6.43% → +HMM 8.41% → +LLM 9.61%. AFM anchors context; HMM specialises; LLM repair beats the quant ceiling.

4 Decay: S&P halved, SSE quartered (Sharpe change)

Round-Trip Friction	Ann. Return	Net Sharpe	Max Drawdown	Calmar Ratio
0 bps (Frictionless)	11.84%	0.7420	-18.21%	0.6502
5 bps	10.73%	0.6892	-18.95%	0.5662
10 bps (Baseline)	**9.61%**	**0.6365**	**-19.68%**	**0.4883**
15 bps	8.50%	0.5731	-20.42%	0.4163
20 bps	7.38%	0.5098	-21.16%	0.3488
30 bps	5.15%	0.3831	-22.65%	0.2274

S&P decay -35.03% to -6.77%. China -92.08% to -46.68%. Repair removes overfit noise instead of deleting trees.

5 What the machine rediscovers (feature mix)



Feature mix of top alphas (paper Fig. 6): intermediate momentum + volatility scalers dominate. Sensible economics: trends in calm markets, reversals + quality in stress.

6 Read this and leave (5-min summary)

- **Robust every year:** positive Sharpe 2019–25 incl. COVID (-14.8% DD vs -38.5% momentum).
- **Deployable:** about $250M capacity; costs in; $0 inference (local 3B/2B models).
- **Next:** NSE 500, LoRA-Gemma, multi-agent judges.

Take-away: semantic repair is a free regulariser, India to US to China.

7 Repair example + wealth curve + regimes (one glance)

Before (1/5): `add(log(vol_5d), sin(p_to_sma20))` — sine on prices is noise. After (4/5): `div(ret_20d, ts_mean(vol_20d,20))` — risk-adjusted momentum. Rank IC −0.012 → +0.038; 100% repairs compile, zero crashes over 1,729 days.

Wealth 2019–25 compounds through bull + bear (momentum DD −40.65%).

HMM: State 0 calm expansion vs State 1 turbulence (COVID-2020, tightening-2022); stickiness 0.994/0.983.

Repair turns nonsense into economics; regime routing is why wealth survives shocks.

Refs: Kakushadze 101 Alphas (2016) ▪ AlphaAgent arXiv:2502.16789 ▪ TEDE (2026) ▪ DEAP JMLR ▪ Hamilton (1989) McLean (2016) ▪ Almgren–Chriss (2000). NSE 50 2002–25; S&P 500 + SSE transfer test. Draft 1 (poster.pdf) kept as full version.